Xunjian Yin

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Duke University, Durham - 27705, US.

Research Interests

#Recursive Self-Improvement #Continual Learning #Long Horizon Agent

Education

- **Duke University** *08.2025 - 12.2028 (expected)*
  Ph.D. in Computer Science, co-advised by Prof. Shuyan Zhou and Prof. Bhuwan Dhingra
- **Peking University** *09.2022 - 06.2025*
  M.S. in Computer Science, supervised by Prof. Xiaojun Wan GPA: 3.81/4.00
- **Peking University** *09.2018 - 06.2022*
  B.S. in Computer Science GPA: 3.64/4.00

Selected Publications

Preprints:

[1] **WeAct: Dynamic Weight Switching as Agent Actions** *arXiv 2026*
Rui Shi*, **Xunjian Yin***, Jiacheng Sang*, Zichuan Liu, Shuyan Zhou.

[2] **From Atomic to Composite: RL Enables Generalization in Complementary Reasoning** *arXiv 2025*
Sitao Cheng, **Xunjian Yin**, Ruiwen Zhou, ..., Liangming Pan, William Yang Wang, Victor Zhong.

Conference paper:

[1] **Constructing Challenging Browser-Use Tasks by Controlled Environment Interventions**
NeurIPS 2026
Xunjian Yin*, Tianchen Guan*, Jinao Wang, ..., Bhuwan Dhingra, Xiangjun Wang, Shuyan Zhou.

[2] **Gödel Agent: A Self-Referential Agents Framework for Recursively Self-Improvement**
ACL 2025
Xunjian Yin, Xinyi Wang, Liangming Pan, Xiaojun Wan, William Wang.

[3] **LEDOM: Reverse Language Model**
ACL 2026
Xunjian Yin, Sitao Cheng, Yuxi Xie, ..., Xinyi Wang, Liangming Pan, William Wang, Xiaojun Wan.

[4] **The Geometry of Reasoning: Flowing Logics in Representation Space**
ICLR 2026
Yufa Zhou*, Yixiao Wang*, **Xunjian Yin***, Shuyan Zhou, Anru R. Zhang.

[5] **DAMON: A Dialogue-Aware MCTS Framework for Jailbreaking Large Language Models**
EMNLP 2025
Xu Zhang, **Xunjian Yin**, Dinghao Jing, Huixuan Zhang, Xinyu Hu, Xiaojun Wan.

[6] **ChemAgent: Self-updating Memories in LLMs Improves Chemical Reasoning**
ICLR 2025
Xiangru Tang, Tianyu Hu, Muyang Ye, Yanjun Shao, **Xunjian Yin**, ..., Arman Cohan, Mark Gerstein

[7] **Benchmarking Knowledge Boundary for LLMs: A Different Perspective on Model Evaluation**
ACL 2024
Xunjian Yin*, Xu Zhang*, Jie Ruan, Xiaojun Wan.

[8] **History Matters: Temporal Knowledge Editing in Large Language Models**
AAAI 2024
Xunjian Yin, Jin Jiang, Liming Yang, Xiaojun Wan.

[9] **ALCUNA: Large Language Models Meet New Knowledge**
EMNLP 2023
Xunjian Yin*, Baizhou Huang*, Xiaojun Wan.

[10] **How Do Seq2Seq Models Perform on End-to-End Data-to-Text Generation?**
ACL 2022
Xunjian Yin, Xiaojun Wan.

[11] **Themis: A Reference-free NLG Evaluation Model with Flexibility and Interpretability**
EMNLP 2024
Xinyu Hu, Li Lin, Mingqi Gao, **Xunjian Yin**, Xiaojun Wan.

Research Experience

- **Google DeepMind** *05.2026 - 09.2026*
  ***Advisor**: Dr. Jingmeng Rao* ***Role**: Student Researcher* Mountain View, USA
  - **Project**: Working Memory Management for Long Horizon Agentic Tasks
    - We designed a method that enables internal agents to autonomously and dynamically manage their context and retrieve necessary information from their history without loss.
- **Amazon AGI SF Lab** *10.2025 - 05.2026*
  ***Manager**: Xiangjun Wang* ***Role**: Research Collaborator* San Francisco, USA
  - **Project**: World Models and Benchmarks for Computer-Use Agents
    - We build world models for computer-use agents and benchmarks to evaluate them.
- **University of California, Santa Barbara (NLP Group)** *06.2024 - 02.2025*
  ***Advisor**: Prof. William Wang* ***Role**: Visiting Research Scholar* California, USA
  - **Project 1**: Reverse Language Model (RLM) Pre-training, Evaluation, Analysis and Applications
    - We trained RLMs (2B-7B) from scratch, using a 500 billion tokens and the last-token prediction approach.
  - **Project 2**: Gödel Agent: A Self-Referential Agents Framework for Recursively Self-Improvement
    - We developed a self-referential agent, which is capable of reading and modifying its own logic and code.
- **Microsoft Research Asia (NLC Group)** *02.2022 - 08.2022*
  ***Advisor**: Dr. Shuming Ma and Dr. Kai Chen* ***Role**: Research Intern* Beijing, China
  - **Project:** Pre-training with Curriculum Learning (CL)
    - We investigated the application of CL during pretraining, which accelerates the convergence of the pretraining.
- **Wangxuan Institute of Computer Technology, Peking University** *10.2020 - 06.2022*
  ***Advisor**: Prof. Xiaojun Wan* ***Role**: Research Assistant* Beijing, China
  - **Project 1**: Analysis of Seq2Seq Models on End-to-End Data-to-Text Generation
    - We find that the most advanced models do not always yield the best performance and larger models are better.
  - **Project 2:** Enhancing Language Models with kNN for Grammar Error Correction (GEC)
    - A kNN algorithm with robust information derived from character phonetics and shapes to augment LMs.
- **Institute of Computational Linguistics, Peking University** *04.2020 - 11.2021*
  ***Advisor**: Prof. Yunfang Wu* ***Role**: Research Assistant* Beijing, China
  - **Project:** Multi-Task Learning for Grammar Error Correction
- **Institute of Computational Linguistics, Peking University** *07.2019 - 12.2019*
  ***Advisor**: Prof. Sujian Li* ***Role**: Research Assistant* Beijing, China
  - **Project:** Building Benchmark for Mathematical Olympiad Problems

Honors and Awards

- **Gold Reviewer Award,** ICML 2026 *05.2026*
- **Outstanding Graduate of Beijing,** Beijing Municipality *06.2025*
- **Outstanding Graduate of Peking University,** Peking University *06.2025*
- **Outstanding Master's Thesis Award of WICT,** Peking University *06.2025*
- **Merit Student,** Peking University *09.2024*
- **Guotai Junan Scholarship,** Peking University *09.2024*
- **Wang Xuan Scholarship,** Peking University *09.2022*
- **Award for Research Excellence,** Peking University *09.2021, 09.2022, 09.2023*
- **Award for Academic Excellence,** Peking University *09.2019, 09.2020*
- **Outstanding Student of Shandong Province,** Shandong Province *04.2018*

Service

- **Teaching Assistant:** Duke University (2026); Peking University (2021 - 2023)
  - Natural Language Processing (2026 spring, with Prof. Bhuwan Dhingra)
  - Web Data Mining (2023 fall, with Prof. Xiaojun Wan)
  - Introduction to Computing (Python, 2023 fall, with Prof. Xifeng Deng)
  - Data Structures and Algorithms (2022 spring, with Prof. Zhao Wang)
  - Introduction to Computing (C++, 2021 fall, with Prof. Yibo Lin)
- **Reviewer:** *2022 - Present*
  - ACL-ARR (2023 - Now), ICLR (2024 - 2026), NeurIPS (2024 - 2026), ICML (2025 - 2026), AAAI (2024 - 2025), COLM (2025), COLING (2024), JCST (2025)
- **Volunteer:** *2023 - Present*
  - AAAI'24, ACL'24, NLPCC'23 Shared Task 8 track chair

Skills

- Pytorch, Transformers, Distributed Training, Accelerated Inference
- Python, C++, HTML, JavaScript